Sajan Senghera

604-612-7739 | sajan.senghera@gmail.com | LinkedIn | Github | Website | Burnaby, B.C.

Technical Skills

Languages: Python, SQL/NoSQL, C++, JavaScript/TypeScript, Bash, HTML, Lua, C#
Frameworks: .NET, Bootstrap, ROS1, Convex, MongoDB, Express.js, Pytorch, Flask, React.js, PostgreSQL, TimescaleDB, Next.js
Libraries: Pandas, Scikit-Learn, Numpy, Scipy, Rospy, Websocket, Axios
Developer Tools: Jupyter, Redis, DevOps, AWS, RESTful APIs, Docker, Git, CI/CD, Jest, Playwright, Linux, Biome

Relevant Experience

Data Scientist Jan 2026 – Present
Clockwork Energy *Waterloo, ON*

- In my Co-op, developed a production root cause analysis engine that diagnoses EV charger anomalies using telemetry from 2,000+ chargers and millions of OCPP events, ranking likely failure causes with recommended resolutions to automate operator troubleshooting.
- Performed large-scale exploratory data analysis on charger telemetry to uncover operational trends, showing that a 1% increase in charging success generated over $1,000 in additional monthly energy dispensed.
- Investigated abnormal heartbeat behavior across millions of events, identifying a software defect responsible for nearly 600 false offline alerts, significantly reducing unnecessary operator interventions.

STEAM & Coding Instructor May 2024 – Oct 2024
Science Alive *Vancouver, BC*

- Authored and delivered technical training workshops and curriculum for **150+** students on advanced Python and full-stack web development concepts.
- Authored and updated **technical documentation**, including standardized lesson plans and debugging guides to ensure curriculum consistency and improve delivery processes.
- Coordinated with fellow instructors and parents to report on student progress and gather feedback, achieving a 95% positive feedback rate.

Research Assistant Jul. 2024 – Dec. 2024
Tangent Lab, SFU *Burnaby, BC*

- Conducted **statistical analysis** on large gesture recognition datasets, defining key performance indicators and achieving 98% accuracy with low latency.
- Developed and optimized complex simulation environments integrating Leap Motion sensor with Z1 robotic arm using **ROS1** in a dockerized environment to reduce setup time by 80%.
- Engineered and deployed a real-time gesture recognition pipeline utilizing a **Pytorch CNN** and integrated the output with a **Q-Learning** module for adaptive robotic arm control.

Projects

Code Royale | *Convex, Next.js, Typescript, Bun* Sep. 2025 – Jan. 2026

- Engineered a **Convex** backend with a **Next.js 15** frontend to manage real-time, high-concurrency multiplayer state for technical interviews, supporting squads and live match launches.
- Developed a full-stack real-time data layer using Convex for state synchronization, enabling a functional **Spectate Mode** that scales to **100+ concurrent users**.

Pokemon Stock Checker | *Python, asyncio, Playwright, Chrome DevTools Protocol* Nov. 2024 – Present

- Built a high-frequency Amazon.ca product monitoring and checkout automation system using asynchronous HTTP polling, Playwright, and Amazon TVSS availability detection.
- Implemented seller and price validation, cart and checkout automation, browser reconnection, event logging, and dashboard tracking for 100+ successful checkout attempts.

Education

Simon Fraser University (SFU) Sep. 2023 – Present
Bachelor of Science in Computing Science *Burnaby, BC*

- SFU Undergraduate Scholars Entrance Scholarship with Distinction recipient
- **Relevant Courses**: Database Systems, Software Engineering, Systems Programming, Artificial Intelligence